

APPLICATIONS

A CLASSIFICATION SCHEME FOR THE CONSISTENCY OF BATSMEN

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Key words: Batting consistency; cricket; moving averages; moving consistency.

Summary: The consistency of batsmen is evaluated by means of the coefficient of variation. The distribution of the coefficients of variation of a large number of batsmen in ODIs is studied and a classification scheme is proposed whereby the consistency of batsmen can be indicated. A graphic procedure is proposed whereby a batsman can be evaluated by looking at his average and consistency jointly. Procedures are also discussed for the monitoring of a batsman's performance on a match-by-match basis.

1. Introduction

When a batsman goes in to bat in a one-day international (ODI) or test match, details are given of his career performance, including the average number of runs he has scored per completed innings. Nothing, however, is indicated about his batting consistency. The purpose of this paper is to propose that a simple measure, the coefficient of variation, be used to classify batsmen into several classes, according to their consistency. Our suggestion is that by looking at the batsman's average and his consistency class, spectators will get a better indication of the quality of the batsman.

The main purpose of this paper is to present something that will be practical, easy to understand and to apply.

2. Graphical assessment of performance

This part of our proposal is closely related to the method proposed by Barr and Van den Honert (1998), who plotted each batsman's average number of runs against the standard deviation of his runs. They point out that the 45 degrees straight line in a plot of $E(R)$ versus $\sigma(R)$, where R denotes the number of runs scored, represents the case where the coefficient of variation, $\sigma(R)/E(R)$, equals 1. This line is seen as the division between consistent and less consistent batsmen. They state that $E(R)/\sigma(R)$ is a measure of consistency of scoring. We prefer to use its inverse, $\sigma(R)/E(R)$, as a more direct and easily understood measure of consistency, and to plot each batsman's average against his coefficient of variation (CV). In this type of plot one can see along the vertical axis what the batsman's average is (as in their paper), and along the horizontal axis how consistent the batsman is: The lower his CV, the more consistent he is. From such a graph it is much easier to compare batsmen's consistencies than from the type of graph proposed by Barr and Van den Honert (1998).

In cricket statistics it is standard practice to calculate a batsman's average based on his scores per completed innings. By completed innings we mean that if a player is not out when an innings ends, his score is carried forward to the next innings and added to his next score, etc., until he gets out. We will adhere to this convention in the calculation of averages.

The use of completed innings scores, however, often lead to unrealistically high scores which are not typical of the player's performance. Such scores

cause the standard deviation to increase dramatically and therefore also the CV. This creates the impression that the player is very inconsistent, which is totally unrealistic. We therefore propose that for the assessment of players' consistency, the **match** scores should be used (irrespective of the question whether the player was out or not). The mean, the standard deviation and the CV are calculated for the match scores. We then plot the average (based on **completed** innings) against the CV (based on match scores). Note that the average based on completed innings equals the total number of runs divided by the number of times the batsman was out.

In figures 1–5 we give such plots for players of the S.A. ODI team until April 1998, March 1999, October 1999, March 2000 and December 2000, respectively. Note that in each case we use a player's full record up to the given date in accordance with the practice regarding players' averages. *The vertical lines at specific CV-values will be explained later.*

3. Classification scheme for consistency of batsmen

As in Nel (1998) we examine the distribution of CVs by considering the 200 ODI players from all countries with the most matches played (at least 25 each) from the start of ODI matches until April 1998. We then propose a classification system whereby batsmen are divided into ten consistency classes by using percentile intervals of the distribution of CVs. The top 10% of batsmen (those with the lowest CV values) fall in class 1, the next 10% in class 2, etc. up to the least consistent 10% who fall in class 10.

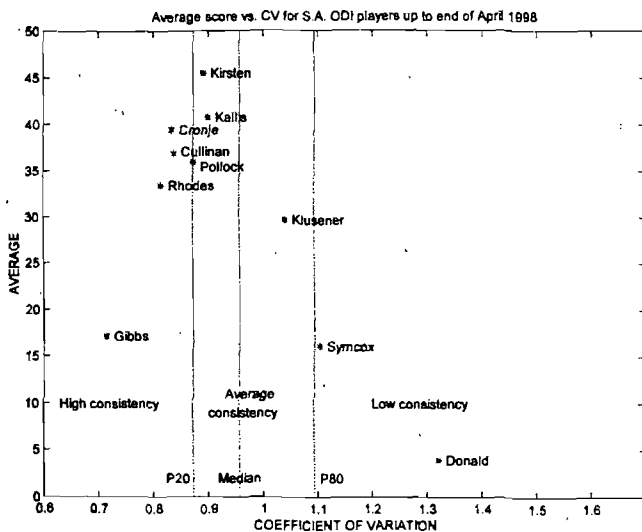


Figure 1

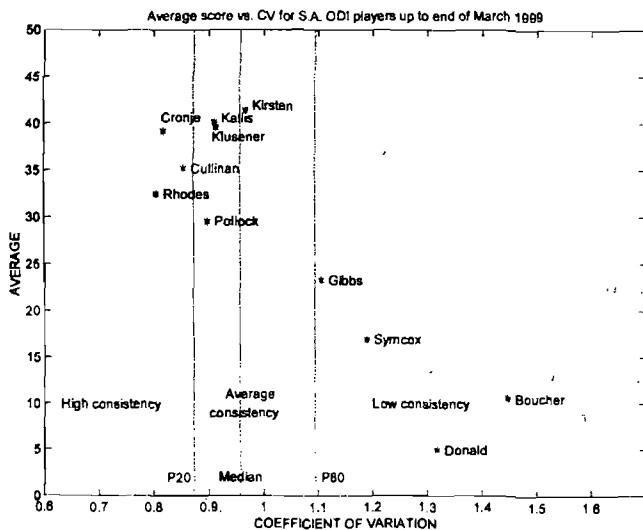


Figure 2

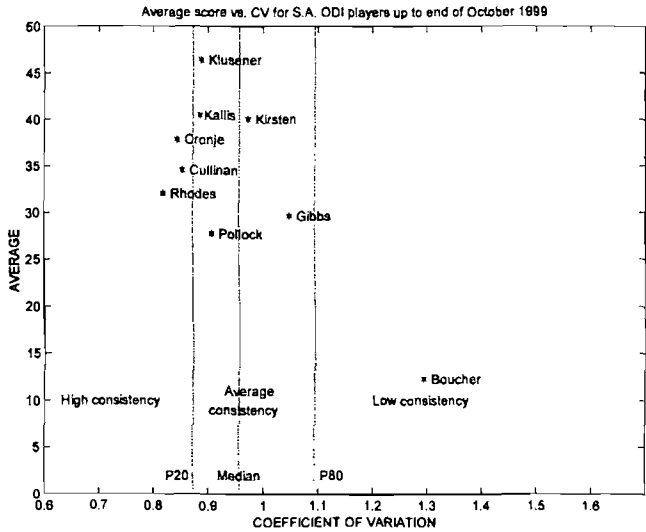


Figure 3

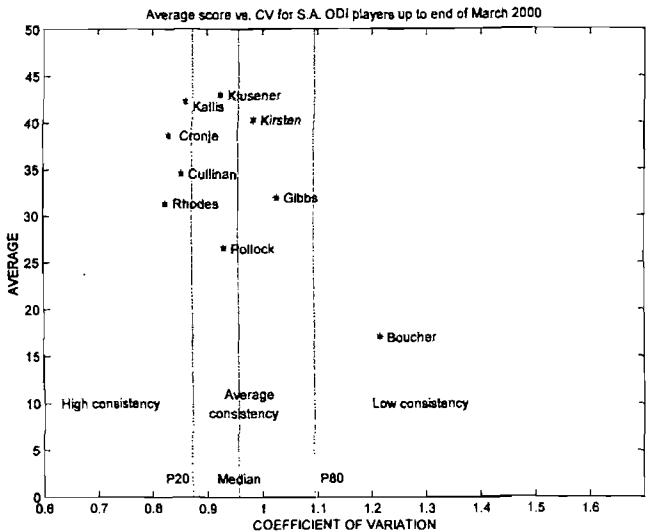


Figure 4

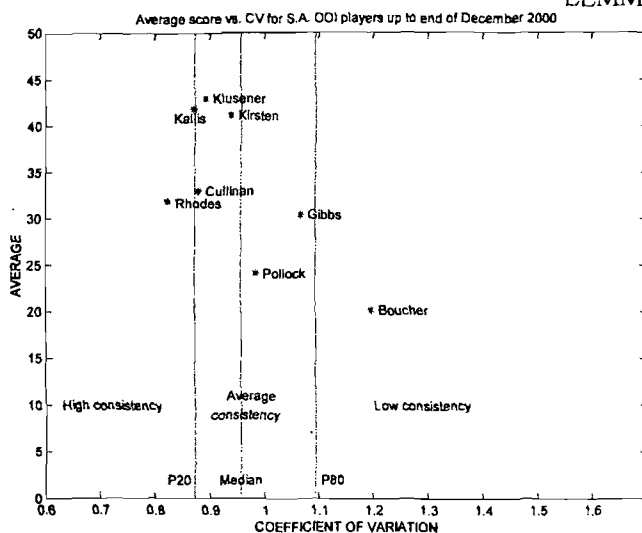


Figure 5

Nel (1998) has fitted various distributions to the CVs (based on completed innings) and concluded that the lognormal distribution gave the best fit. By transforming the percentiles of the normal distribution to the lognormal distribution, Nel obtained his classification scheme – cf. Nel (1998) table 1. These percentiles depend on the adequacy of the lognormal distribution.

We have calculated the CVs of the same 200 players by using the match scores. Attempts to get a distribution that fits the data well, failed: The lognormal fit was even worse than Nel's fit.

An important question is, however, whether the data really follow a specific simple distribution. Having at our disposal such a large data set (16 544 scores) which reflects all the different kinds of peculiarities of ODIs under all kinds of circumstances, we have decided to base our classification scheme on the data set itself: We use the percentiles of the data set. We have also used the bootstrap technique to simulate the percentiles and these agreed very well with the empirical percentiles.

The classification scheme is given in table 1.

Table 1

<u>Consistency interval</u>	<u>Consistency class</u>
$0 - 0,835$	1
$0,835^+ - 0,873$	2
$0,873^+ - 0,898$	3
$0,898^+ - 0,931$	4
$0,931^+ - 0,957$	5
$0,957^+ - 0,983$	6
$0,983^+ - 1,027$	7
$1,027^+ - 1,094$	8
$1,094^+ - 1,201$	9
$1,201^+ - \infty$	10

The intervals in table 1 are data driven, but this is after all a classification scheme and it can be used as such for a long time without the need of updating. Actually, unless the rules of the game are drastically modified, it would not be advisable to update it frequently.

We suggest that when a batsman goes in to bat, his average score and consistency class should be given. In table 2 we give the averages, CV values, consistency classes (denoted CI) and n , the numbers of innings played by a group of South African players based on performances up to five different dates, which facilitates the monitoring of players' performances over consecutive periods. See also figures 1–5 where the players closest to the upper left-hand corner of a figure are the best in terms of average and consistency jointly. The three vertical lines, P_{20} , median and P_{80} denote the 20th, 50th and 80th percentiles. These classify the players as having high, average and low batting consistency respectively.

We notice that Cronjé and Rhodes remained very consistent through all periods, and also Cullinan over 4 periods. These three players maintained a

fairly stable average throughout. Looking at Kirsten we notice that a decrease in his average from 45.47 to 41.53 has been accompanied by a decrease in his consistency from class 3 to class 6, and although his average remained fairly constant thereafter, his consistency did not improve, which was mainly due to occasional high scores. Klusener's average increased rapidly and then stabilised, and his consistency has also stabilised.

Pollock's average as well as his consistency decreased steadily. Kallis' average fluctuated slightly but his consistency improved.

By looking at a batsman's average and consistency class **jointly**, one can judge his performance. A low class number (high consistency) indicates that his match scores remained approximately the same.

It is very interesting to look at a batsman's consistency on a continuous basis. In figures 6 and 7 we plot the consistency of Kirsten and Rhodes after their first twenty innings and then after each innings thereafter on a cumulative basis: Each point represents his career consistency up to that specific innings. The 20th, 50th and 80th percentiles are given, according to which consistency is judged. The letters on the horizontal axis denote the ends of certain months (seasons), namely a: April 1994, b: April 1996, c: April 1997, d: April 1998, e: March 1999 and f: March 2000. Looking at figure 7 we notice that Rhodes was extremely consistent up to his 78th score. In his 79th innings he scored 121 runs; which caused the sudden leap in his CV. Kirsten, on the other hand, has scored ten centuries fairly evenly over most of his career. Being an opening batsman, he is in a better position than Rhodes to score centuries. Both these players are excellent batsmen, despite the differences in their averages and consistencies. See also figures A1–A6 in the appendix for other players.

Table 2

Player	April 1998				March 1999			
	Ave	CV	CI	n	Ave	CV	CI	n
Cronjé	39.42	0.834	1	127	39.23	0.817	1	148
Rhodes	33.41	0.815	1	121	32.50	0.805	1	137
Cullinan	36.88	0.838	2	90	35.19	0.855	2	109
Kirsten	45.47	0.891	3	87	41.47	0.967	6	103
Pollock	35.95	0.873	2	37	29.56	0.898	3	51
Klusener	29.73	1.040	8	29	39.61	0.913	4	44
Kallis	40.76	0.899	4	42	40.13	0.910	4	63
Gibbs	17.18	0.715	1	17	23.31	1.106	9	27
Boucher					10.58	1.447	10	23

Player	October 1999				March 2000			
	Ave	CV	CI	n	Ave	CV	CI	n
Cronjé	37.86	0.844	2	160	38.65	0.831	1	175
Rhodes	32.03	0.818	1	148	31.35	0.824	1	153
Cullinan	34.56	0.853	2	118	34.56	0.853	2	118
Kirsten	40.04	0.973	6	112	40.33	0.984	7	124
Pollock	27.74	0.907	4	60	26.52	0.931	4	70
Klusener	46.43	0.887	3	56	42.96	0.923	4	72
Kallis	40.48	0.886	3	75	42.26	0.860	2	89
Gibbs	29.65	1.048	8	39	32.00	1.026	7	56
Boucher	12.23	1.295	10	34	17.03	1.214	10	46

Player	November 2000			
	Ave	CV	CI	n
Cronjé				
Rhodes	31.84	0.824	1	171
Cullinan	32.99	0.879	3	133
Kirsten	41.13	0.937	5	148
Pollock	24.11	0.984	7	81
Klusener	42.97	0.891	3	91
Kallis	41.85	0.870	2	113
Gibbs	30.42	1.067	8	59
Boucher	20.20	1.197	9	59

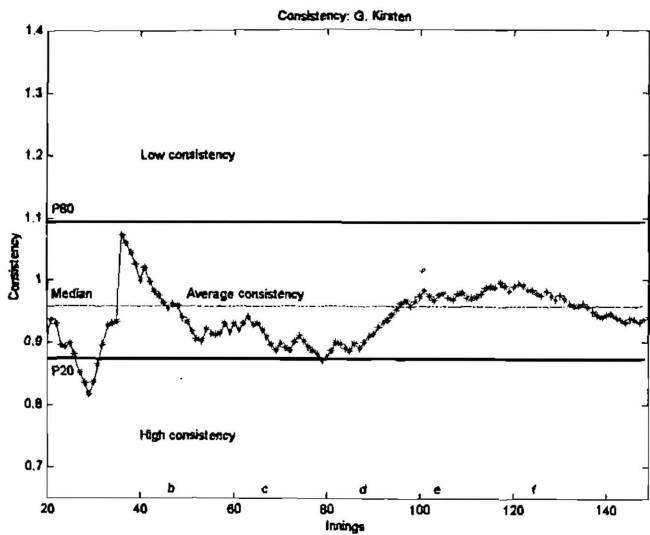


Figure 6

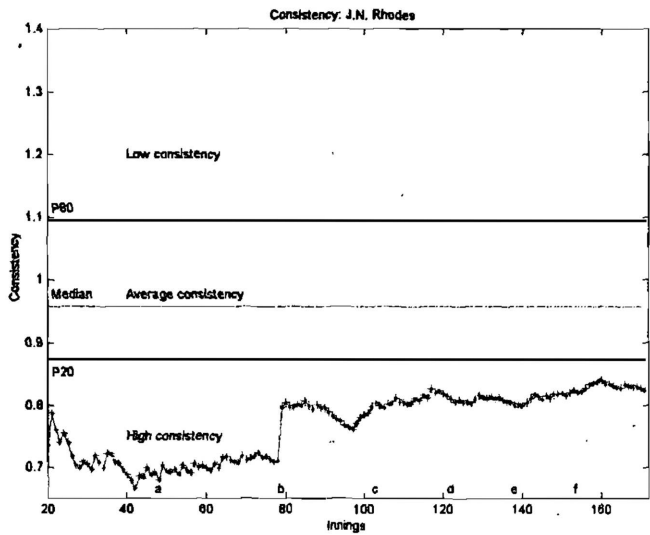


Figure 7

We believe that the methods proposed are simple but informative enough to convince cricket authorities to use them. By utilising the computer, which is already used to calculate the averages, to calculate the *CVs* also, a further dimension can be added to the enjoyment of the spectator. Moreover, a useful further criterion can be given to selectors and trainers in assessing players' performance.

For cricket authorities' purposes updating can be done on a regular basis, but for comparison purposes it would be interesting to compare players' performance from season to season, or after completion of a test series.

4. The use of moving averages and consistencies

If we want to monitor a player's performance on a continuous basis, various graphic methods can be used. We can start by plotting the player's individual match scores against the numbers of the innings. This normally gives a very erratic picture. We could do some smoothing by using a moving window and calculating a moving average within the window – cf. Hart (1997) §2.2. This, however, is a form of curve fitting and is not what cricket authorities would find useful. For their purposes a procedure based on the ideas of control charts would be more suitable in order to identify a drastic change in a player's performance. To evaluate a player's present form it is logical to look at his most recent scores: Not only the last one, but rather the last group consisting of, say, r scores. This motivates the use of moving averages. After every innings, we calculate the average of the r most recent scores. We now plot each moving average against the number of the last innings – cf. Grant and Leavenworth (1980) pp. 293–298 for more details. The present form of the player can be judged from the last few points. In order to have a yardstick to

measure whether the player performs normally, we can calculate control limits, as in the case of control charts. Consecutive averages are, however, dependent, so those points outside the control limits have to be judged carefully – cf. Grant and Leavenworth (1980) pp. 295–296. Although some assumptions in the deduction of the control limits are not met, the limits still serve as bounds that can be used to take action.

We now use data up to December 2000 to illustrate the procedure. Denote the true scores of a player by $x_1, x_2, \dots, x_n$. The central line of the control chart is determined by $\bar{x}$, the average of the player's ODI scores. According to Montgomery (1991) p. 307 the upper control limit is given by $UCL = \bar{x} + 3\sigma/\sqrt{r}$. We estimate σ by s/c_4 (p. 232) where s denotes the standard deviation of the scores. Thus $UCL = \bar{x} + 3s/\sqrt{r}c_4$ and $LCL = \bar{x} - 3s/\sqrt{r}c_4$. We found the 3-sigma control limits very wide, and will rather use 1.96-sigma limits.

The choice of the value of r is subjective. We have studied the moving averages of various players by using $r = 3, 5$ and 7 . For $r = 3$ they are very erratic, becoming smoother for $r = 5$ and $r = 7$. From a practical point of view, if action has to be taken based on the average of a player's last games, we believe that a value of $r = 5$ is a good choice. In figures 8 and 9 we give plots of moving averages for Rhodes and Kirsten. See also figures A7–A12 in the appendix. Our suggestion is that the control limits can be used to judge a player's performance over the next season's matches and that updating (recalculation of control limits) can be done annually.

Batsmen's consistencies can also be monitored on a moving five-point basis. The determination of upper and lower bounds for individual batsmen is, however, unfeasible because the distribution of the CVs is unknown.

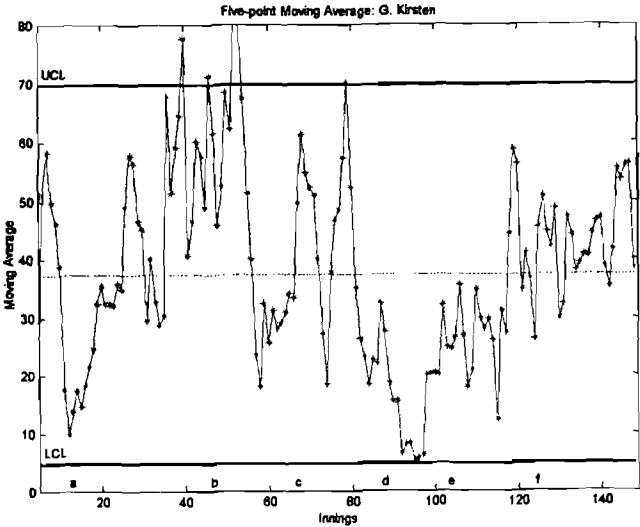


Figure 8

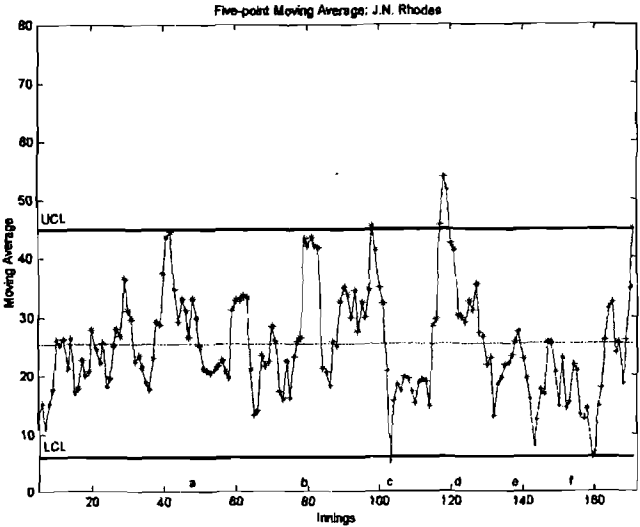


Figure 9

We rather use general bounds determined in a similar fashion to those in section 3. We cannot use the same bounds because we now have CVs calculated for samples of size five each. We used the same data set as before and selected a random sample of size 50 from the 200 players. For each one we calculated the CV of each moving sample of size five. We now use the percentiles from this data set (consisting of 4155 samples). The percentiles are given in the following table:

Percentage:	10	20	30	40	50	60	70	80	90
Percentile:	0.55	0.64	0.72	0.79	0.87	0.95	1.04	1.14	1.30

The moving consistency of Kirsten and Rhodes are given in figures 10 and 11 (and of other players in figures A13–A18 in the appendix). In this case each point is based on exactly five consecutive scores. Only the points in the region of low consistency (above $P80$) are of interest: Examine whether a high CV was due to one or more high scores. If so, this does not reflect negatively against the player. By using the moving averages and moving consistencies jointly, selectors can judge the present form of a batsman. This can be of great value to trainers and selectors.

5. Concluding remarks

1. In our two-dimensional assessment of players' career performances (figures 1–5 or table 2) we used firstly the conventional averages, based on completed innings. This is standard practice. Secondly we use the very logical and simple measure of dispersion, namely the CV based on match scores. We believe that such a two-dimensional plot or table is simple to understand and also very informative, especially if it is updated from time to time.

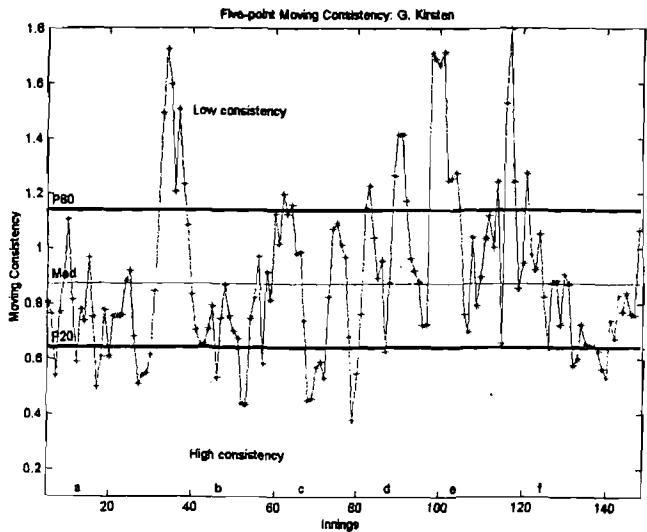


Figure 10

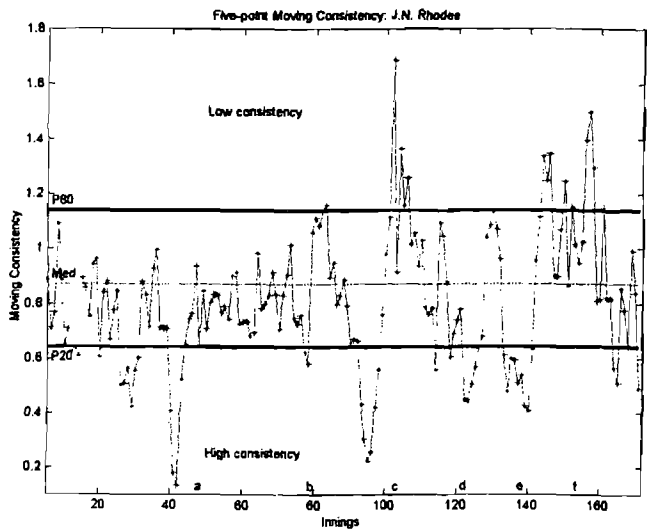


Figure 11

2. The five-point moving averages and consistencies are objective instruments which quantify players' performances on a continuous basis and can be very valuable for selectors.
3. Barr and Van den Honert (1998) use a measure of consistency to adjust a batsman's average, thereby arriving at one figure, the so-called 'adjusted average' which, on its own, does not seem to say much. They mention that B. McMillan's consistency-adjusted average is 47.80 compared to his average of 39.52, but from a statistical point of view there is no justification to use the adjusted average as an estimator for his future performance. We, on the other hand, find it much more informative to give a two-dimensional evaluation of a batsman, either by plotting his average against his CV in a graphic display, or otherwise by stating his average and consistency class.
4. The exclusion of players who have played fewer than 25 matches might have shortened the right-hand tail of the distribution, but some line had to be drawn in order to ensure that players who are still very inexperienced, are excluded. Taking into account that some players had over 200 matches, the minimum of 25 seemed reasonable.
5. We considered it advisable to use a large data base (e.g. players of all ODI countries over a long period of time) in order to get a stable and representative standard of evaluation. Updating the classification intervals occasionally, but not too frequently, may be advisable.
6. A consistency table for ordinary test matches can be drawn up using the same technique as for ODIs.
7. Instead of five-term equally weighted moving averages one can use

exponentially weighted moving averages which place the largest weight on the last score and progressively smaller weights on scores further back in time – cf. Montgomery (1991) pp. 299–300 and 309. Results in this connection can be obtained from the first author.

8. Almond et al. (2000) present an interesting graphical method which could also be used to compare players by means of a display of confidence intervals.
9. Preston and Thomas (2000) used dynamic programming to determine the best batting strategy in limited overs cricket.

Acknowledgements

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APPENDIX

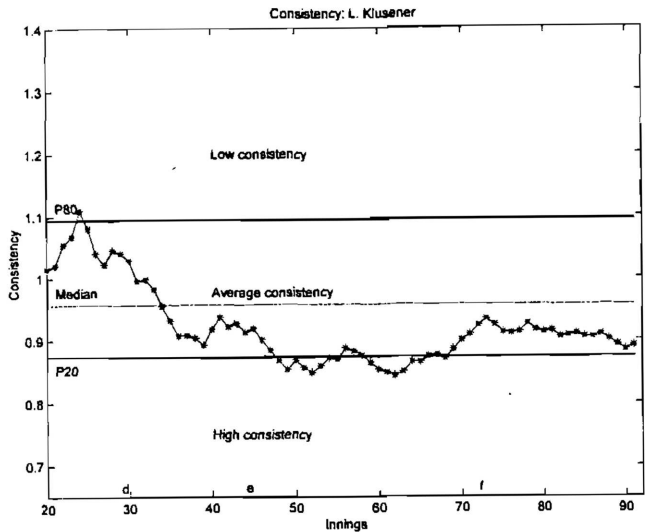


Figure A1

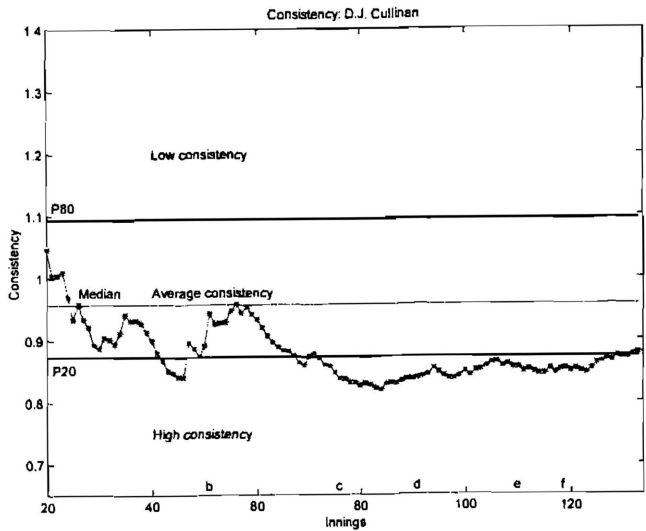


Figure A2

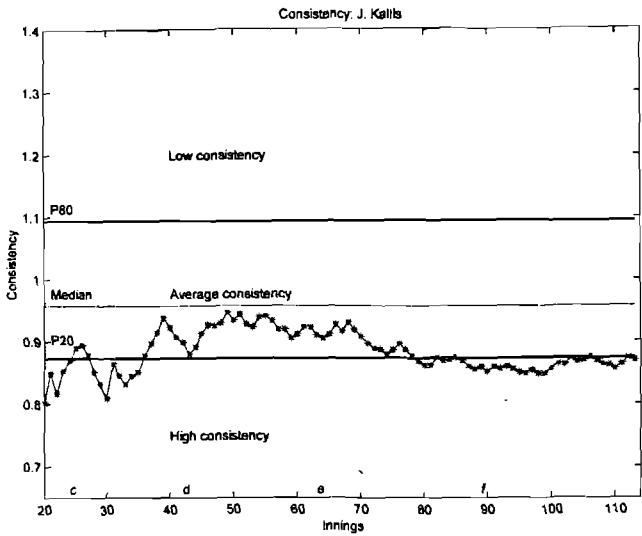


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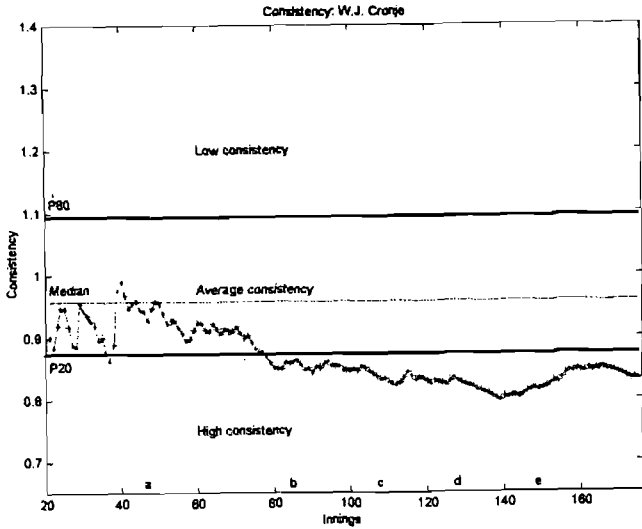


Figure A4

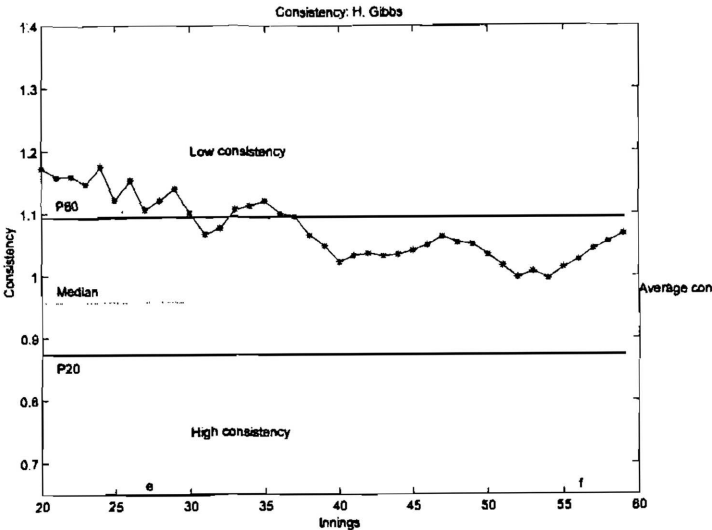


Figure A5

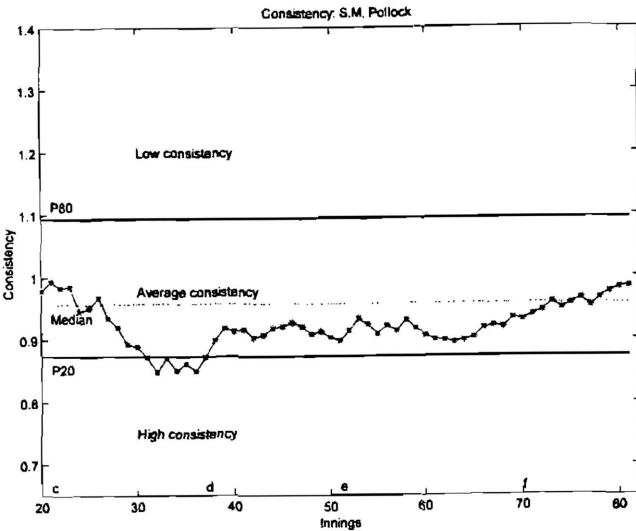


Figure A6

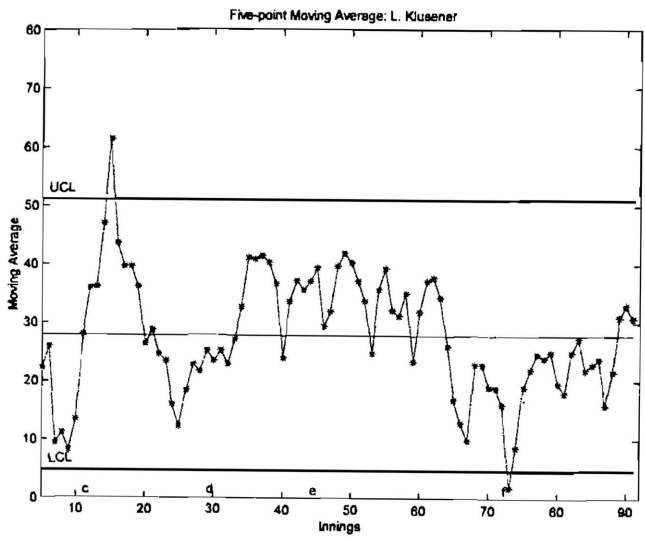


Figure A7

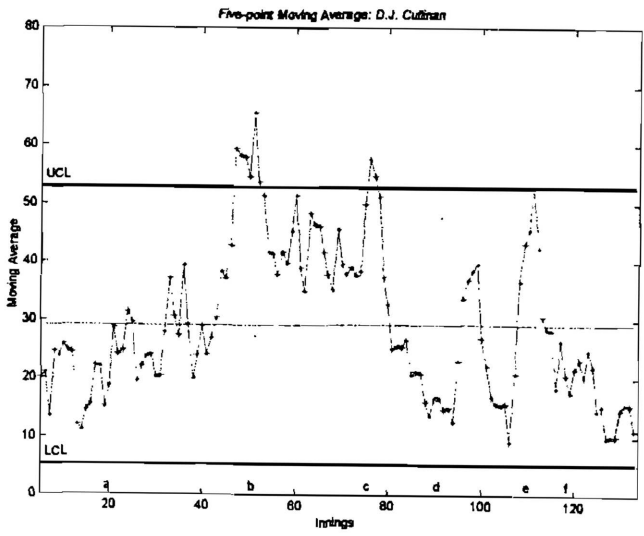


Figure A8

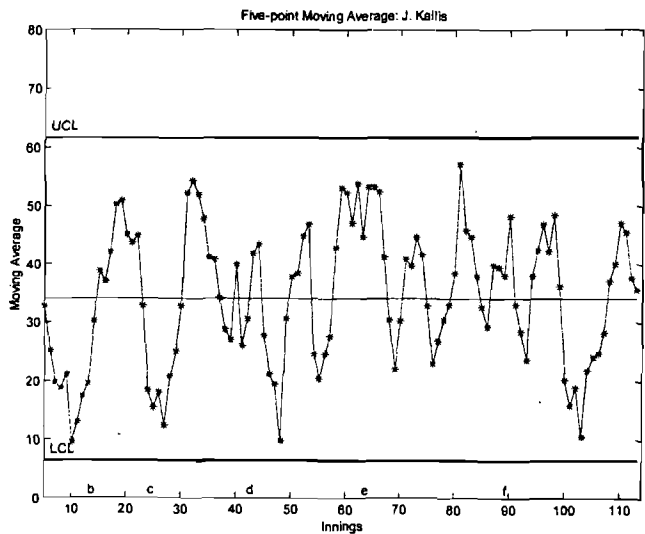


Figure A9

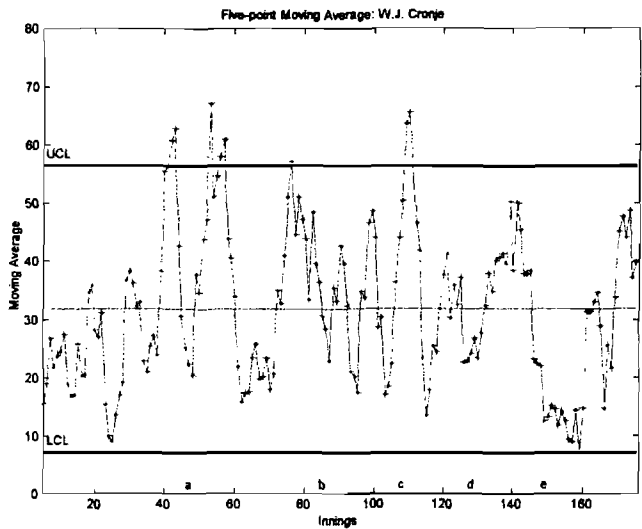


Figure A10

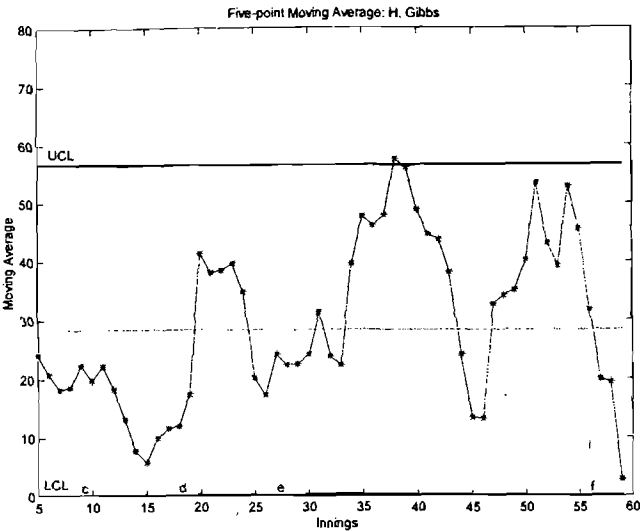


Figure A11

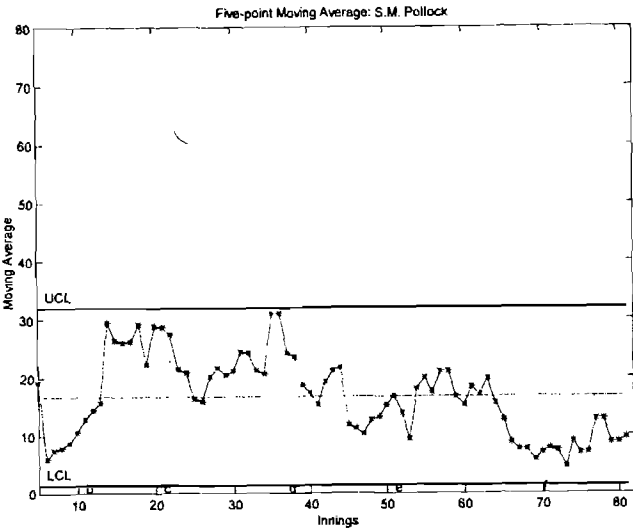


Figure A12

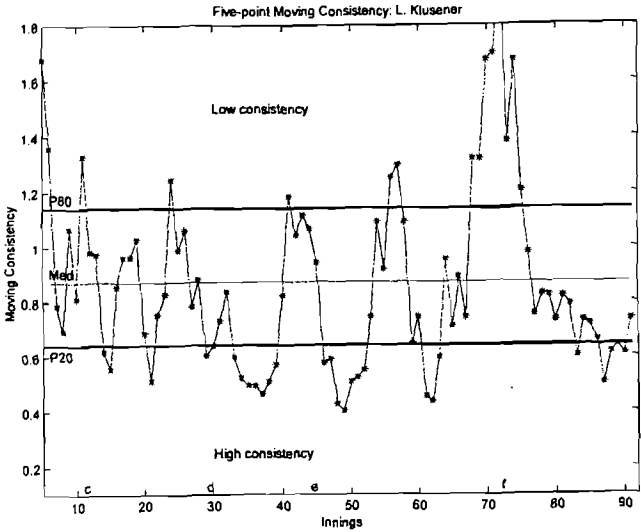


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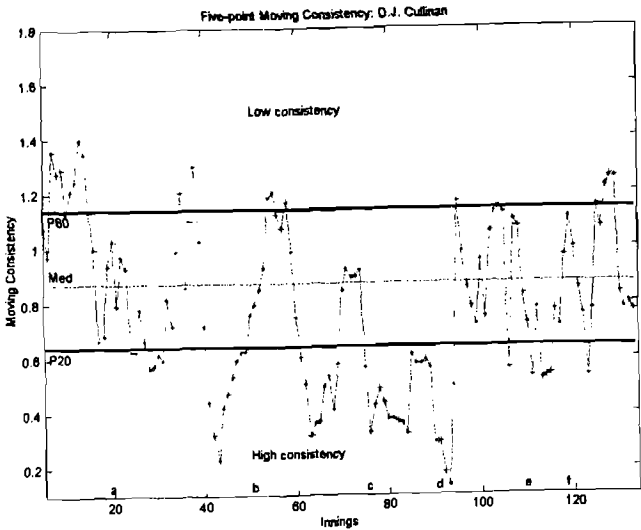


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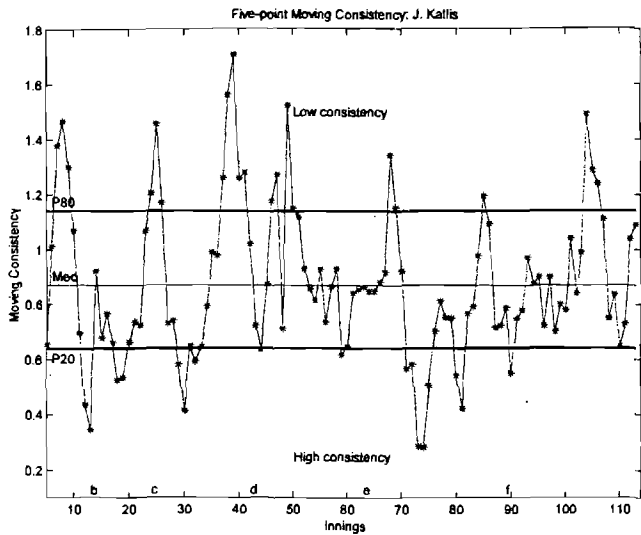


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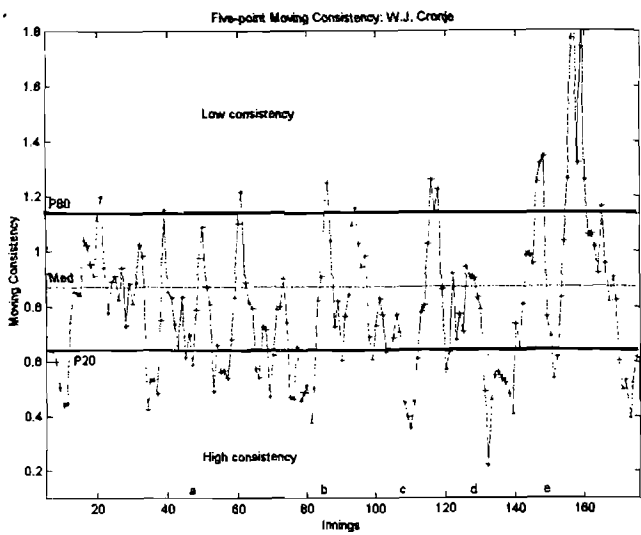


Figure A16

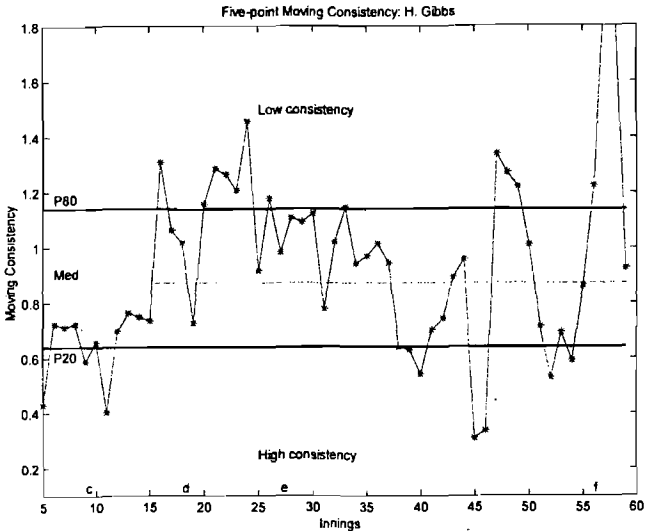


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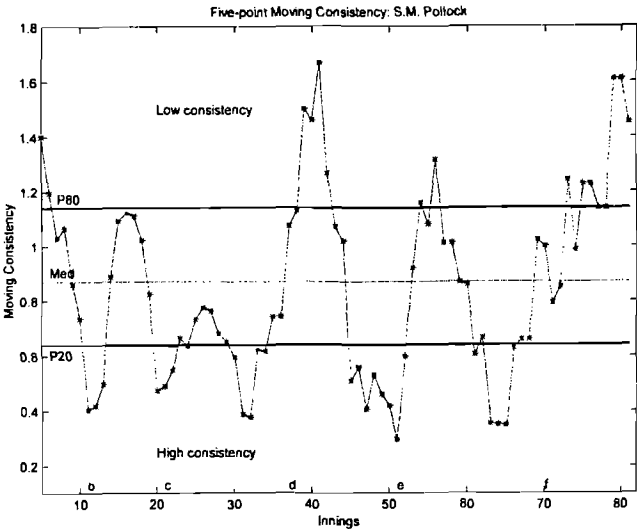


Figure A18